

Capítulo 5

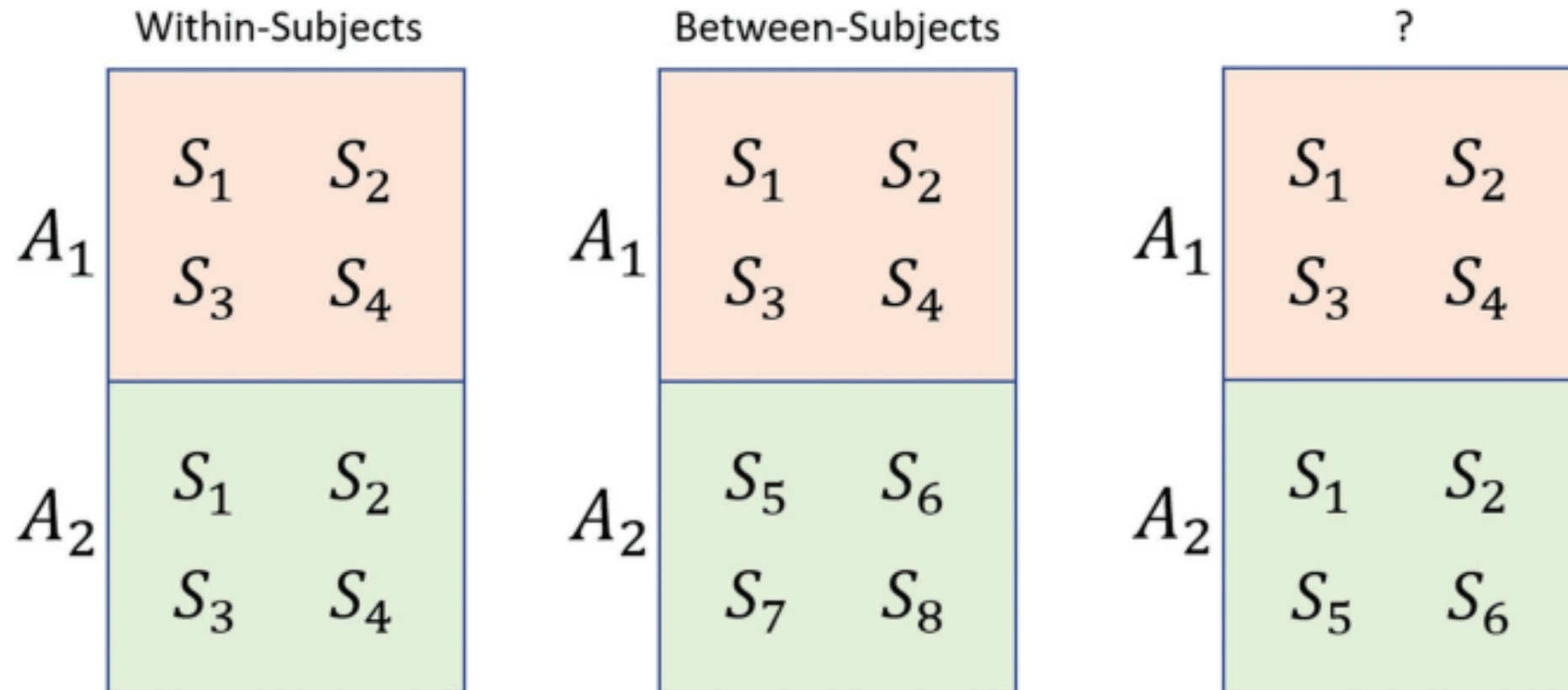
Chapter Precap

- Comparar dos grupos e introducir el concepto de factores entre y dentro de los sujetos.
- Se presenta la codificación de factores y se compara la codificación de 'sum coding'.
- Discutir la descomposición de la variación de la variable dependiente en efectos asociados a variables independientes.
- Discutir la recuperación y manipulación de las muestras posteriores del modelo.
- Se introduce la función de hipótesis y se explica la recuperación y combinación de parámetros de "efecto aleatorio".
- Finalmente, se discuten los valores atípicos y la robustez, y se presenta la distribución t.

Comparing Two Groups

- Observamos bajo dos condiciones diferentes.
- Podemos preguntarnos: ¿La distribución de las observaciones bajo una condición es la misma que la distribución bajo la otra condición?
- Por lo general, nos enfocamos en: ¿La media es igual para ambas condiciones?, es decir, $\mu_{A_1} = \mu_{A_2}$ para las condiciones A_1 y A_2 ?

Between and Within Subjects



Data

```
# load packages and data
library (bmmb)
library (brms)
data (exp_data)
# exclude actual men and apparent men
notmen = exp_data[exp_data$C_v != 'm' & exp_data$C != 'm',]
```

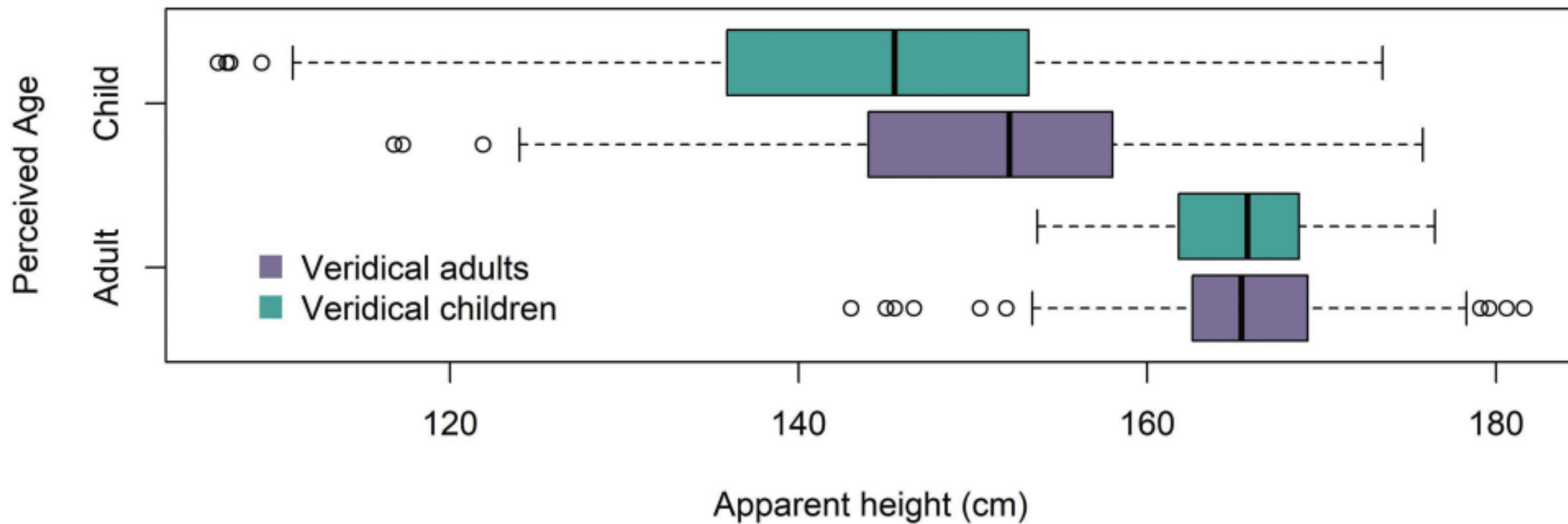
- **L**: An integer from 1 to 15 indicating which *listener* responded to the trial.
- **height**: A floating-point number representing the *height* (in centimeters) reported for the speaker on each trial.
- **S**: An integer from 1 to 139 indicating which *speaker* produced the trial stimulus.
- **A**: The *apparent age* of the speaker indicated by the listener, **a** (adult) or **c** (child).

Confusable Voices

```
xtabs ( ~ bmmb::exp_data$C_v + bmmb::exp_data$C)
##               bmmb::exp_data$C
## bmmb::exp_data$C_v    b     g     m     w
##                   b 234 133     6   32
##                   g  79 184     0   22
##                   m  31   0 626   18
##                   w  97 109     3 511
```

```
xtabs ( ~ notmen$A_v + notmen$C)
##               notmen$C
## notmen$A_v    b     g     w
##           a  97 109 511
##           c 313 317  54
```

Apparent Age and Height



Research Questions

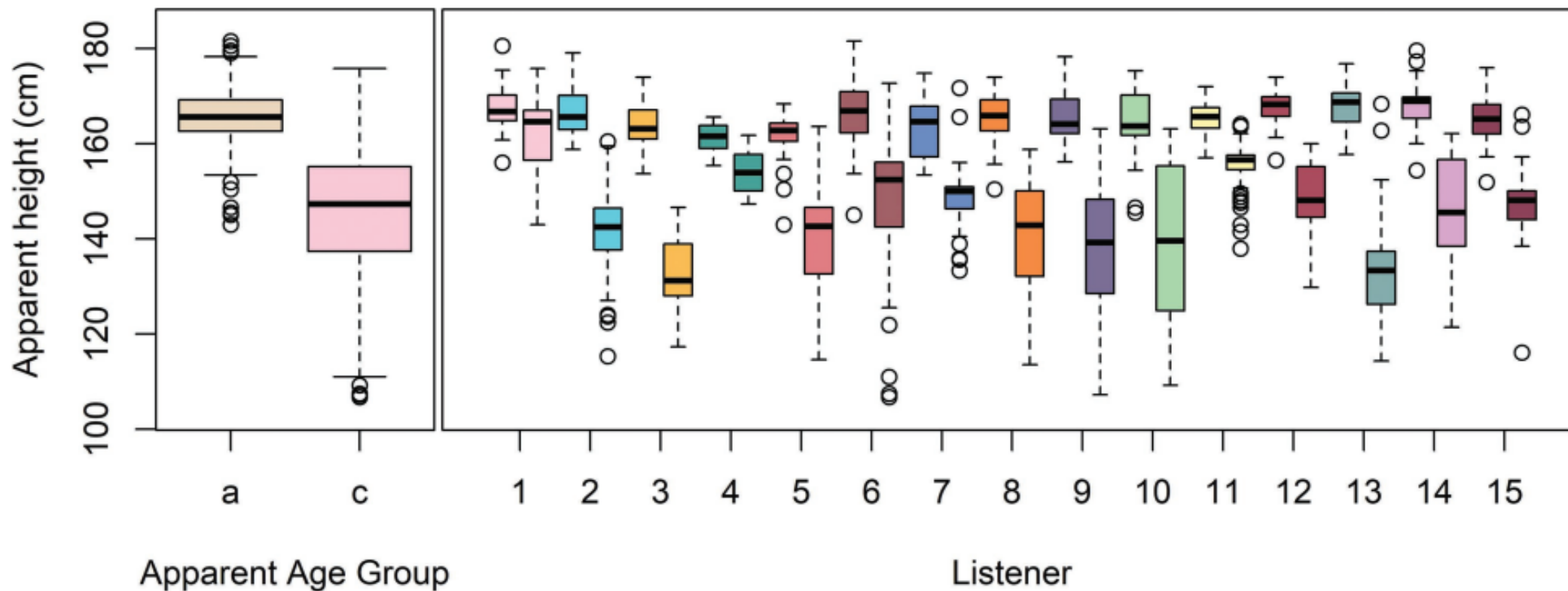
- (1) ¿Qué tan altos suenan los hablantes percibidos como mujeres adultas?
- (2) ¿Qué tan altos suenan los hablantes percibidos como niños?
- (3) ¿Cuál es la diferencia de altura aparente asociada a la percepción de la adultez?

Between and Within

```
xtabs (~ notmen$A + notmen$L)
##           notmen$L
## notmen$A  1  2  3  4  5  6  7  8  9 10 11 12 13 14 15
##           a 42 35 42 20 34 37 30 48 42 46 42 44 31 36 36
##           c 52 59 50 74 58 55 64 46 49 48 52 50 63 58 58
```

```
xtabs (~ notmen$A + notmen$S) [,40:50]
##           notmen$S
## notmen$A 40 41 42 43 44 45 46 92 93 94 95
##           a 11  0  0  0  0  0  0 11  7 15 14
##           c  4 15 15 15 15 15 15  4  8  0  1
```

Differences Between Two Groups



The Model Formula

- Nuestro último modelo contenía solo un efecto "fijo" de intercepto.

```
height ~ 1 + (1|L) + (1|S)
```

- Para incluir la edad aparente (A), simplemente lo añadimos fuera de los paréntesis:

```
height ~ 1 + A + (1|L) + (1|S)
```

- Podemos omitir el '1' cuando hay otro predictor presente:

```
height ~ A + (1|L) + (1|S)
```

Fitting the Model

- `Intercept`: this is a unique class, only for intercepts.
-  `b`: This class includes all fixed-effect predictors *apart* from the intercept.
- `sd`: This is for standard deviation parameters related to ‘batches’ of parameters, e.g. `sd(Intercept)` for $L(\sigma_L)$.
- `sigma`: the error term.

```
# Fit the model yourself
model = brms::brm (
  height ~ A + (1|L) + (1|S), data = notmen, chains = 4, cores = 4,
  warmup = 1000, iter = 3500, thin = 2,
  prior = c(brms::set_prior("normal(156, 12)", class = "Intercept"),
 brms::set_prior("normal(0, 12)", class = "b"),
    brms::set_prior("normal(0, 12)", class = "sd"),
    brms::set_prior("normal(0, 12)", class = "sigma")))
```

Interpreting the Model

```
# inspect model
bmmb::short_summary (model)
## Formula:  height ~ A + (1 | L) + (1 | S)
##
## Group-Level Effects:
## ~L (Number of levels: 15)
##           Estimate Est.Error l-95% CI u-95% CI
## sd(Intercept)    5.24     1.11     3.54     7.85
##
## ~S (Number of levels: 94)
##           Estimate Est.Error l-95% CI u-95% CI
## sd(Intercept)    3.59     0.43     2.8     4.51
##
## Population-Level Effects:
##           Estimate Est.Error l-95% CI u-95% CI
## Intercept    163.92     1.46    160.97    166.72
## Ac           -17.37     0.74    -18.81    -15.89
##
```



```
# calculate mean apparent height based on apparent adultness
tapply (notmen$height, notmen$A, mean)
##      a      c
## 165.5 145.4
```

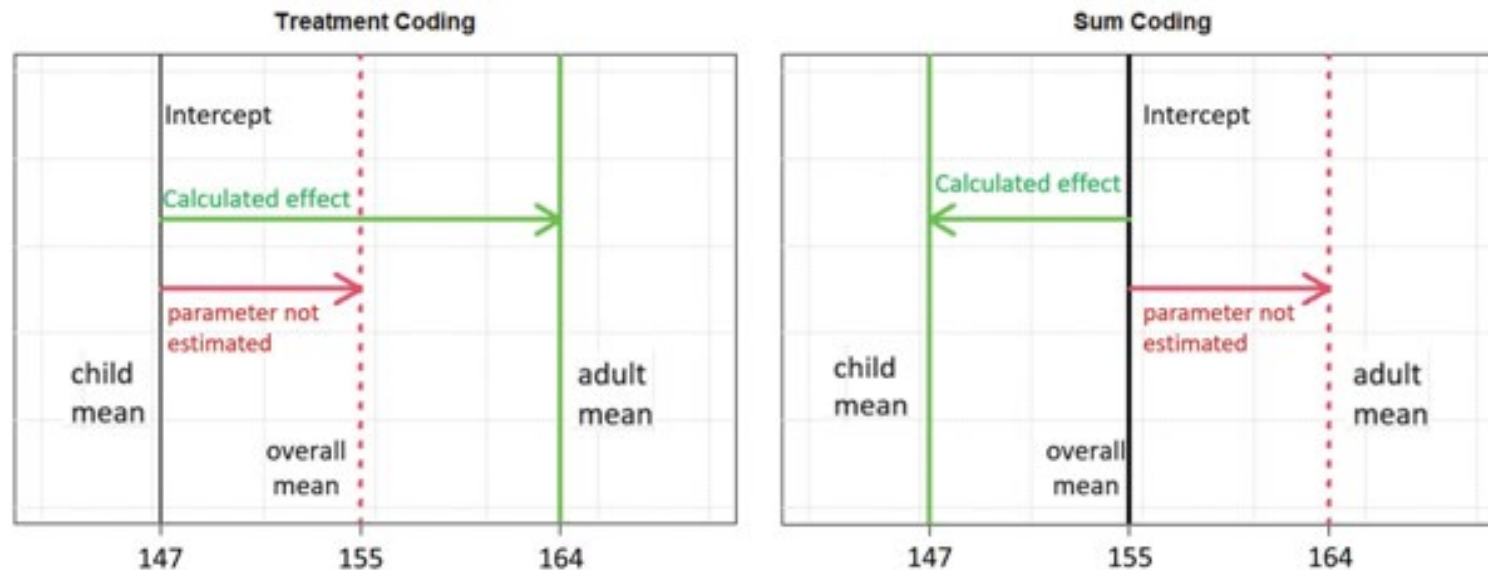
Contrasts

- Su modelo debe elegir una 'referencia'.
- Todos los parámetros de factores codifican variación relativa a esta referencia.
- La implementación matemática de esto se denomina 'codificación de contraste'.

Treatment Coding

To interpret treatment-coded coefficients in a regression model:

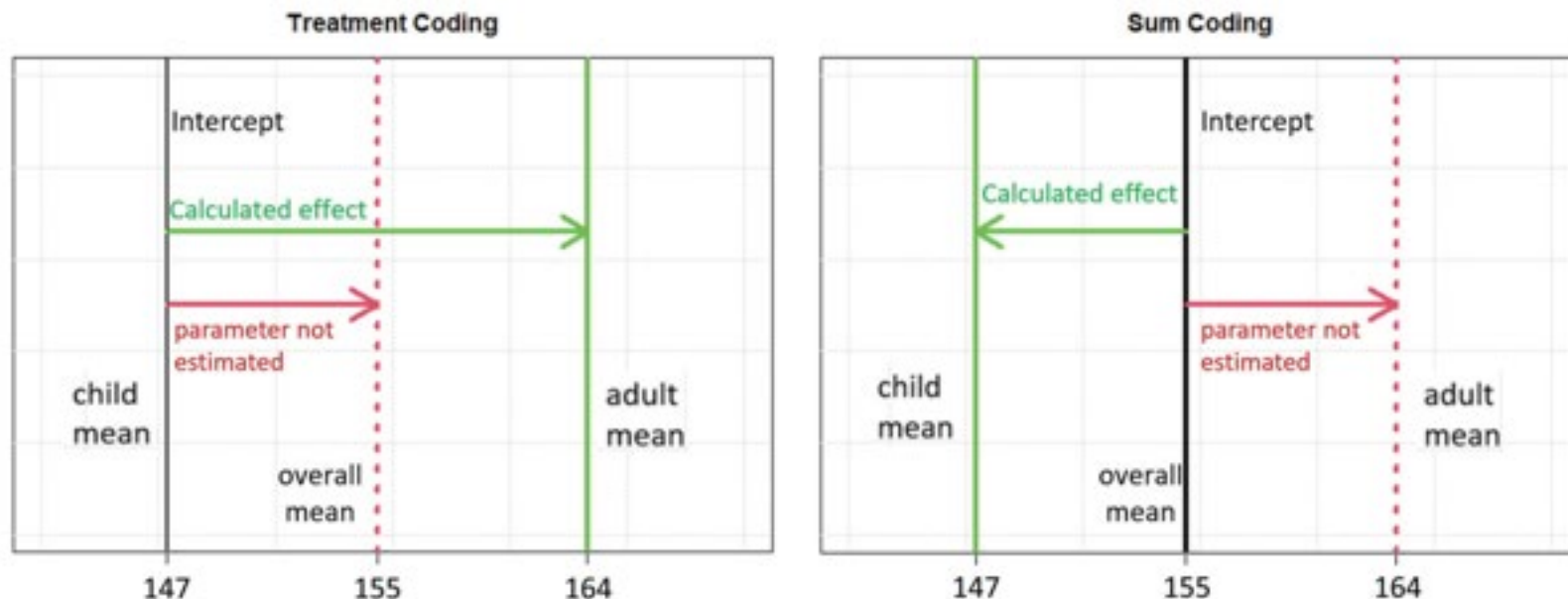
- The reference category mean is the 'Intercept' in the model.
- The value of the coefficients of any non-intercept group is equal to `group mean - Intercept (reference group mean)`.
- To recover the mean estimate for any non-intercept group, we add `group effect + Intercept (reference group mean)`.



Sum Coding

To interpret sum-coded coefficients in regression models:

- The mean of all your group means (the grand mean) is the 'Intercept' in the model.
- The value of the coefficients of any other group mean will be equal to $\text{group mean} - \text{Intercept (grand mean)}$.
- To recover the mean estimate for any other group, we add $\text{group effect} + \text{Intercept (grand mean)}$.



Description of Our Model

¡Observe que la codificación no se representa aquí!


$$\text{height}_{[i]} \sim \mathcal{N}(\mu_{[i]}, \sigma)$$
$$\mu_{[i]} = \text{Intercept} + A + L_{[L[i]]} + S_{[S[i]]}$$

Apparent speaker height is expected to vary according to a normal distribution with some unknown mean (μ) and standard deviation (σ). Means are expected to vary based on whether the listener identified the speaker as an adult or a child (A), and listener and speaker-dependent deviations from the mean (L, S). The listener and speaker effects were modeled as coming from a normal distribution with means of 0 and unknown standard deviations (σ_L, σ_S). The intercept was given a normal prior with a mean of 156 and a standard deviation of 12, and the remaining parameters were given normal priors centered at 0 with standard deviations of 12.

Priors :

$$L_{[\cdot]} \sim \mathcal{N}(0, \sigma_L)$$

$$S_{[\cdot]} \sim \mathcal{N}(0, \sigma_S)$$

$$\text{Intercept} \sim \mathcal{N}(156, 12)$$

$$A \sim \mathcal{N}(0, 12)$$

$$\sigma \sim \mathcal{N}(0, 12)$$

$$\sigma_L \sim \mathcal{N}(0, 12)$$

$$\sigma_S \sim \mathcal{N}(0, 12)$$

Sum Coding and Decomposing Variation

$$\text{height}_{[i]} \sim N(\mu_{[i]}, \sigma)$$

$$\mu_{[i]} = \text{Intercept} + A + L_{[L[i]]} + S_{[S[i]]}$$

- Represent group means as deviations from the intercept.
- Represent the listener and speaker-specific deviations from the intercept ($L_{[\cdot]}, S_{[\cdot]}$) as being centered at 0, with standard deviations of σ_L and σ_S .
- Represent the random error (ϵ) as having a mean of 0 and a standard deviation of σ .
- When a group coefficient is 0 the group lies exactly at the intercept. In sum coding this is the grand mean (the mean of the means) indicating that the group is basically average.
- When a listener or speaker effect is 0 this listener/speaker is exactly average with respect to their group. This means there is nothing about this speaker's average that is unpredictable given knowledge of their group.
- When an error is 0 this production is exactly as expected for a given listener/speaker in a given group. This means that an observation contains no error since it was *exactly* predictable.

Priors :

$$L_{[\cdot]} \sim N(0, \sigma_L)$$

$$S_{[\cdot]} \sim N(0, \sigma_S)$$

$$\text{Intercept} \sim N(156, 12)$$

$$A \sim N(0, 12)$$

$$\sigma \sim N(0, 12)$$

$$\sigma_L \sim N(0, 12)$$

$$\sigma_S \sim N(0, 12)$$

Sum Coding and Decomposing Variation

If we think of our predictors as representing deviations from some reference value, we can 'break up' any observed value into its component parts. For example, suppose that:

- The overall mean is 157 cm.
- The adult female mean is 165 cm (+8 over the intercept).
- A particular speaker has a mean apparent height of 170 cm (+5 over the adult female mean).

If we observe an apparent height judgment of 173 cm for this speaker, that suggests the following decomposition:

$$173 = 157 \text{ (Intercept)} + 8 \text{ (adult female effect)} + 5 \text{ (speaker effect)} + 3 \text{ (error)}$$

This reflects the following considerations:

- The average apparent height across the groups is 157 cm.
- The average for adult females is 8 cm above the overall mean ($157 + 8 = 165$).
- This speaker's average apparent height is 5 cm above the average for adult females ($157 + 8 + 5 = 170$).
- This particular production is 3 cm higher than expected for this particular speaker ($157 + 8 + 5 + 3 = 173$).

Fitting the Model

```
# to change to sum coding  
options (contrasts = c('contr.sum', 'contr.sum'))
```

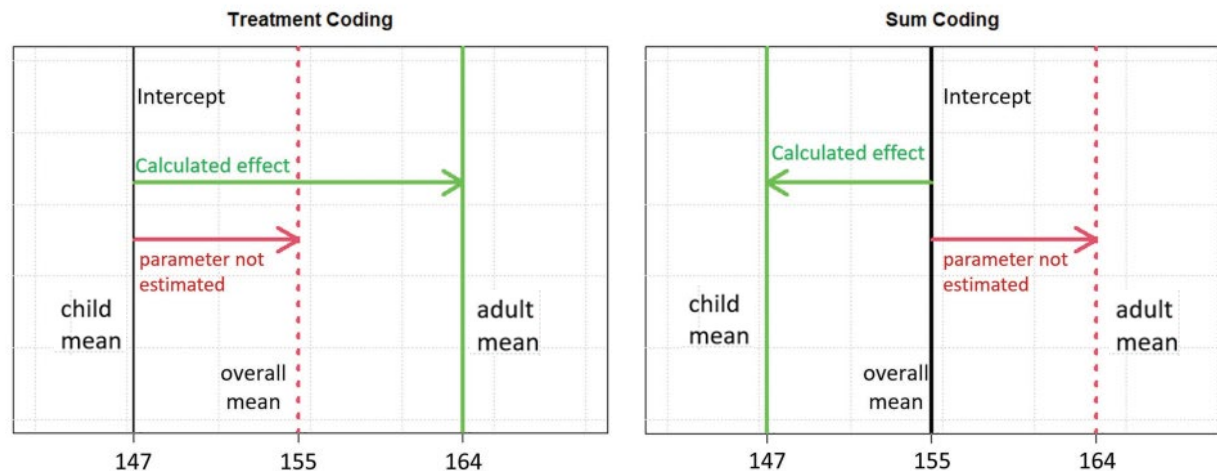
```
# Fit the model yourself  
model_sum_coding = brms::brm (  
  height ~ A + (1|L) + (1|S), data = notmen, chains = 4, cores = 4,  
  warmup = 1000, iter = 3500, thin = 2,  
  prior = c(brms::set_prior("normal(156, 12)", class = "Intercept"),  
            brms::set_prior("normal(0, 12)", class = "b"),  
            brms::set_prior("normal(0, 12)", class = "sd"),  
            brms::set_prior("normal(0, 12)", class = "sigma")))
```

```
# Or download it from the GitHub page:  
model_sum_coding = bmb::get_model ('5_model_sum_coding.RDS')
```


Comparison of Sum and Treatment Coding

```
# sum coding
brms::fixef (model_sum_coding)
##           Estimate Est.Error   Q2.5   Q97.5
## Intercept  155.218    1.4050 152.434 157.965
## A1          8.711     0.3602  8.003   9.416
```

```
# treatment coding
brms::fixef (model)
##           Estimate Est.Error   Q2.5   Q97.5
## Intercept  163.92     1.4580 160.97 166.72
## Ac        -17.37     0.7391 -18.81 -15.89
```



Variance of the Sum of Variables

$$\sigma_{x+y}^2 = \sigma_x^2 + \sigma_y^2 + 2\rho\sigma_x\sigma_y$$

$$\sigma_{x-y}^2 = \sigma_x^2 - \sigma_y^2 - 2\rho\sigma_x\sigma_y$$

Hay que tener en cuenta la correlación.

```
# 'marginal' variance of x
var (x)
## [1] 9.167

# 'marginal' variance of y
var (y)
## [1] 8.773

# variance of sum of x and y
var (x+y)
## [1] 0.6093

# variance of difference of x and y
var (x-y)
## [1] 35.27
```

Combine Before Summarizing

- Las muestras deben combinarse/transformarse antes de resumirlas.
- Resumir y luego combinar puede dar como resultado una estimación de intervalo creíble (varianza) totalmente incorrecta.
- La combinación de muestras posteriores individuales conserva esta información.

```
samples = brms::fixef (model_sum_coding, summary = FALSE)
head (samples)
##      variable
## draw Intercept    A1
##    1      153.2 9.040
##    2      154.7 8.365
##    3      152.2 8.835
##    4      156.0 9.003
##    5      154.7 8.196
##    6      155.9 8.305
```

Bueno

`mean(Intercept + A1)`

Malo

`mean(Intercept) + mean(A1)`

The Hypothesis Function

- Las hipótesis son combinaciones de muestras de parámetros individuales.
- Puedes usar `bmmb::get_samples` para obtener las muestras individuales de sus hipótesis.

Modelo  Hipótesis 

```
bmmb::short_hypothesis(model_sum_coding, "Intercept + A1 = 0")
##      Estimate Est.Error  Q2.5 Q97.5      hypothesis
## H1      163.9      1.465 161.1 166.8 (Intercept+A1) = 0

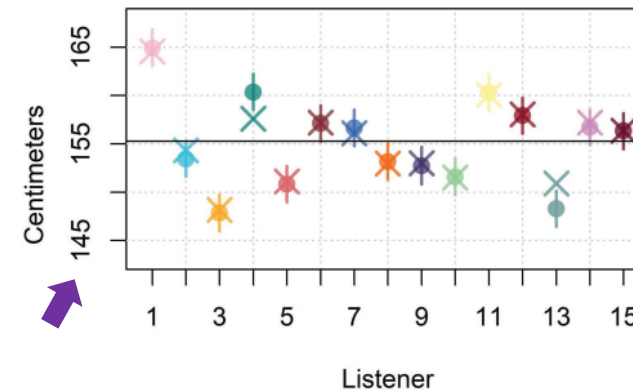
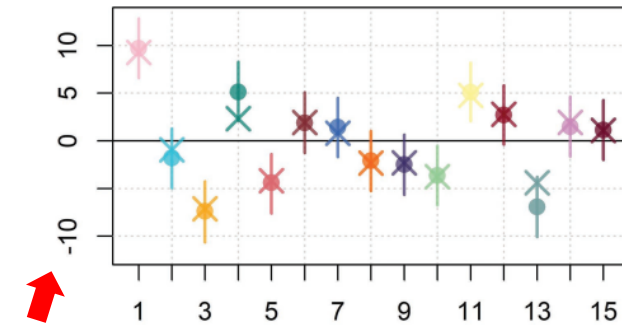
short_hypothesis(model_sum_coding,
                  c("Intercept = 0",      # overall mean
                    "Intercept + A1 = 0",  # adult mean
                    "Intercept - A1 = 0")) # child mean
##      Estimate Est.Error  Q2.5 Q97.5      hypothesis
## H1      155.2      1.405 152.4 158.0      (Intercept) = 0
## H2      163.9      1.465 161.1 166.8 (Intercept+A1) = 0
## H3      146.5      1.435 143.7 149.3 (Intercept-A1) = 0
```

Using Hypothesis for Random Effects

```
short_hypothesis(model_sum_coding, "Intercept = 0")
##      Estimate Est.Error  Q2.5  Q97.5      hypothesis
## H1      155.2      1.405 152.4   158 (Intercept) = 0
```

```
short_hypothesis(model_sum_coding, "Intercept = 0",
                  scope = "ranef", group="L") [1:5,]
##      Estimate Est.Error  Q2.5  Q97.5      hypothesis group
## H1      9.690      1.585   6.603 12.832 (Intercept) = 0      1
## H2     -1.792      1.597  -4.941  1.297 (Intercept) = 0      2
## H3     -7.376      1.602 -10.636 -4.255 (Intercept) = 0      3
## H4      5.121      1.566   2.104  8.283 (Intercept) = 0      4
## H5     -4.393      1.570  -7.598 -1.383 (Intercept) = 0      5
```

```
short_hypothesis(model_sum_coding, "Intercept = 0",
                  scope = "coef", group="L") [1:5,]
##      Estimate Est.Error  Q2.5  Q97.5      hypothesis group
## H1     164.9      0.9516 163.1 166.8 (Intercept) = 0      1
## H2     153.4      0.9438 151.6 155.3 (Intercept) = 0      2
## H3     147.8      0.9585 146.0 149.8 (Intercept) = 0      3
## H4     160.3      0.9556 158.5 162.2 (Intercept) = 0      4
## H5     150.8      0.9475 149.0 152.7 (Intercept) = 0      5
```



Robustness

- Ferrari vs. Honda, ¿cuál es 'mejor'?
- Un Honda es más "robusto": fiable y útil en una gama más amplia de situaciones.
- Robustez distributiva: Asumir menos, asumir cosas más "realistas".

The (non-standardized) t-distribution

- Tres parámetros: Media, escala y parámetro nu (ν , grados de libertad).
- La varianza de la distribución t está determinada por ν .
- El parámetro 'scale' aumenta la varianza (que está totalmente determinada por nu).

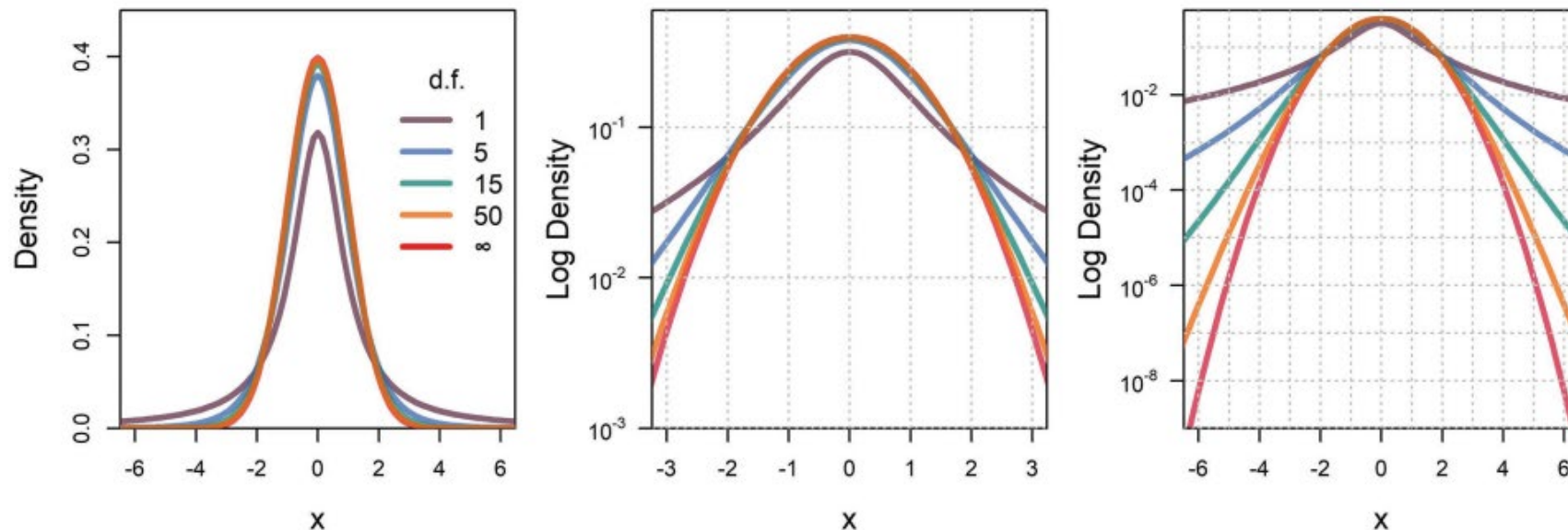
$$t(\nu, \mu_{[i]}, \sigma)$$

$$\sigma = s \cdot \sqrt{\nu / (\nu - 2)}$$

$$\sigma^2 = s^2 \cdot \nu / (\nu - 2)$$

The (non-standardized) t-distribution

- No determina la "normalidad" de la distribución.
- Un nu más pequeño resulta en una distribución más puntiaguda con 'colas gordas', es decir, más valores atípicos.



Outliers

```
resids = residuals (model_sum_coding) [,1]
```

```
range (scale(resids))
```

```
## [1] -4.144  3.471
```

```
head (sort(scale(resids)))
```

```
## [1] -4.144 -3.846 -3.785 -3.679 -3.668 -3.613
```

```
mu = mean(resids)
```

```
sigma = sd(resids)
```

```
# probability of value smaller than smallest outlier
```

```
pnorm (min (resids),mu,sigma)
```

```
## [1] 1.704e-05
```

```
# sample size before outlier this big expected
```

```
1/pnorm (min (resids),mu,sigma)
```

```
## [1] 58680
```


Outliers

```
# get maximum likelihood estimates of t parameters
# the 'lower' bounds are for the sd and df respectively
tparams = MASS::fitdistr (resids, 't', lower = c(0,1))
# check out mean, scale and nu. bottom row is standard errors
tparams
##           m           s           df
##  0.2902    7.1651    7.5753

m = tparams[[1]][1]
s = tparams[[1]][2]
df = tparams[[1]][3]

# probability of value smaller than smallest outlier
bmmb::ptns (min (resids),m, s, df)
##           m
## 0.0007384

# sample size before outlier thie big expected
1/bmmb::ptns (min (resids),m, s, df)
##           m
## 1354
```

Model with t-Distributed Errors

- Puede ser casi exactamente igual que un modelo con errores distribuidos normalmente.
- El parámetro ν (ν) se puede estimar a partir de los datos.

$$\text{height}_{[i]} \sim t(\nu, \mu_{[i]}, \sigma)$$

$$\mu_{[i]} = \text{Intercept} + A + L_{[L[i]]} + S_{[S[i]]}$$

Priors :

$$L_{[\cdot]} \sim N(0, \sigma_L)$$

$$S_{[\cdot]} \sim N(0, \sigma_S)$$

$$\text{Intercept} \sim t(3, 156, 12)$$

$$A \sim t(3, 0, 12)$$

$$\sigma \sim t(3, 0, 12)$$

$$\sigma_L \sim t(3, 0, 12)$$

$$\sigma_S \sim t(3, 0, 12)$$

$$\nu \sim \text{gamma}(2, 0.1)$$

$$\text{height}_{[i]} \sim N(\mu_{[i]}, \sigma)$$

$$\mu_{[i]} = \text{Intercept} + A + L_{[L[i]]} + S_{[S[i]]}$$

Priors :

$$L_{[\cdot]} \sim N(0, \sigma_L)$$

$$S_{[\cdot]} \sim N(0, \sigma_S)$$

$$\text{Intercept} \sim N(156, 12)$$

$$A \sim N(0, 12)$$

$$\sigma \sim N(0, 12)$$

$$\sigma_L \sim N(0, 12)$$

$$\sigma_S \sim N(0, 12)$$

Model with t-Distributed Errors

$$\text{height}_{[i]} \sim t(\overset{\text{red arrow}}{\nu}, \overset{\text{green arrow}}{\mu_{[i]}}, \sigma)$$
$$\mu_{[i]} = \text{Intercept} + A + L_{[L[i]]} + S_{[S[i]]}$$

```
# Fit the model yourself
options (contrasts = c('contr.sum','contr.sum'))
model_sum_coding_t = brms::brm (
  height ~ A + (1|L) + (1|S), data = notmen, chains = 4,
  cores = 4, warmup = 1000, iter = 3500, thin = 2, family="student",
  prior = c(brms::set_prior("student_t(3, 156, 12)", class = "Intercept"),
    brms::set_prior("student_t(3, 0, 12)", class = "b"),
    brms::set_prior("student_t(3, 0, 12)", class = "sd"),
    brms::set_prior("gamma(2, 0.1)", class = "nu"),
    brms::set_prior("student_t(3, 0, 12)", class = "sigma"))
# Or download it from the GitHub page:
model_sum_coding_t = bmmmb::get_model ('5_model_sum_coding_t.RDS')
```

Priors:

$$L_{[\cdot]} \sim N(0, \sigma_L)$$

$$S_{[\cdot]} \sim N(0, \sigma_S)$$

$$\text{Intercept} \sim t(3, 156, 12)$$

$$A \sim t(3, 0, 12)$$

$$\sigma \sim t(3, 0, 12)$$

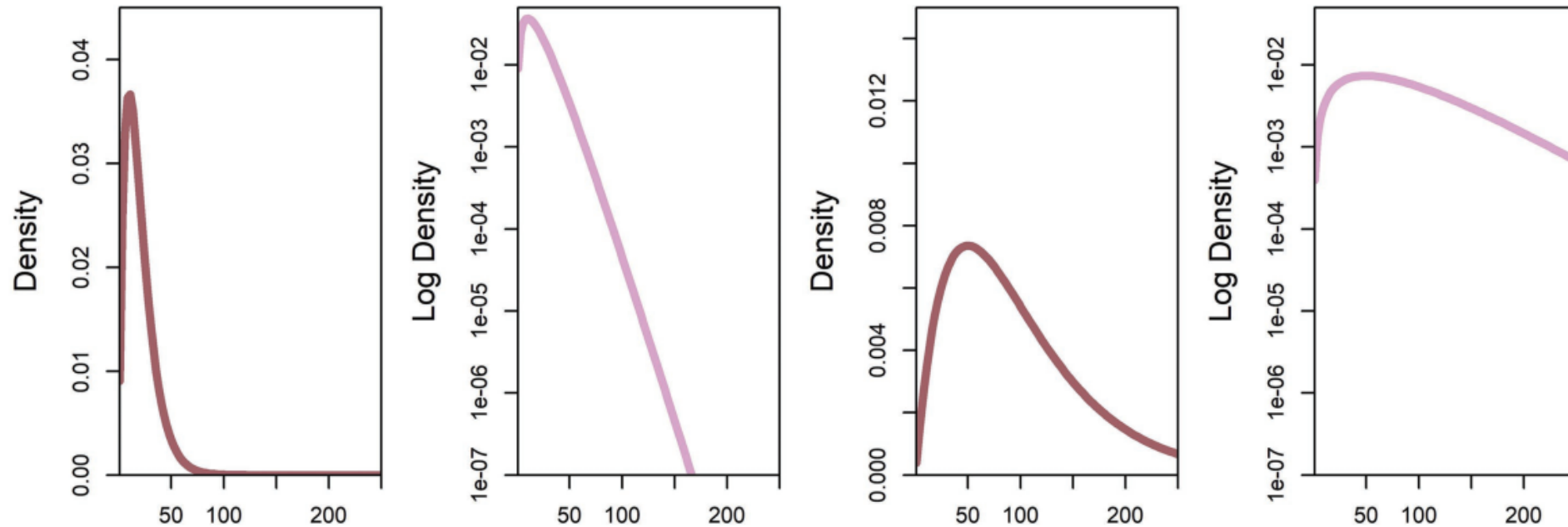
$$\sigma_L \sim t(3, 0, 12)$$

$$\sigma_S \sim t(3, 0, 12)$$

$$\nu \sim \text{gamma}(2, 0.1)$$

Strange Priors

- Fíjate en las formas de las densidades, a ver si éstas "tienen sentido".
- No te preocupes tanto por las matemáticas subyacentes per se.



Inspecting the Model

```
# inspect model
bmb::short_summary (model_sum_coding_t)
## Formula: height ~ A + (1 | L) + (1 | S)
##
## Group-Level Effects:
## ~L (Number of levels: 15)
##
##           Estimate Est.Error 1-95% CI u-95% CI
## sd(Intercept)      5.08      1.12    3.46    7.81
##
## ~S (Number of levels: 94)
##
##           Estimate Est.Error 1-95% CI u-95% CI
## sd(Intercept)      3.36      0.42    2.6    4.21
##
## Population-Level Effects:
##           Estimate Est.Error 1-95% CI u-95% CI
## Intercept      155.59      1.38   152.85   158.31
## A1              8.44      0.34    7.78    9.12
##
## Family Specific Parameters:
##           Estimate Est.Error 1-95% CI u-95% CI
## sigma ←       7.23      0.27    6.71    7.76
## nu ←          6.90      1.52    4.72   10.40
```

$$\text{height}_{[i]} \sim t(\overset{\text{green arrow}}{\nu}, \overset{\text{red arrow}}{\mu_{[i]}}, \sigma)$$
$$\mu_{[i]} = \text{Intercept} + A + L_{[L[i]]} + S_{[S[i]]}$$

Priors:

$$L_{[\cdot]} \sim N(0, \sigma_L)$$

$$S_{[\cdot]} \sim N(0, \sigma_S)$$

$$\text{Intercept} \sim t(3, 156, 12)$$

$$A \sim t(3, 0, 12)$$

$$\sigma \sim t(3, 0, 12)$$

$$\sigma_L \sim t(3, 0, 12)$$

$$\sigma_S \sim t(3, 0, 12)$$

$$\nu \sim \text{gamma}(2, 0.1)$$

Comparing Models

```
fixef (model_sum_coding)
##           Estimate Est.Error      Q2.5      Q97.5
## Intercept  155.218      1.4050  152.434  157.965
## A1          8.711       0.3602   8.003   9.416
fixef (model_sum_coding_t)
##           Estimate Est.Error      Q2.5      Q97.5
## Intercept  155.588      1.3803  152.85  158.311
## A1          8.443       0.3386   7.78   9.119
```


Simulating Two-Group Data

```
n_listeners = 15
n_speakers = 94 # must be even!

# don't run this line if you want a new simulated dataset.
set.seed(1)
```

```
# this is the value of our intercept
Intercept = 155
```

```
# this is a vector of adultness fixed effects
A_ = c(8.7, -8.7)
```

```
# this is a vector indicating the adultness group
A = rep(1:2, (n_listeners*n_speakers/2))
```

```
# this is a vector of 15 listener effects
L_ = rnorm(n_listeners, 0, 5.2)
```

```
# this is a vector indicating the listener
L = rep(1:n_listeners, each = n_speakers)
```

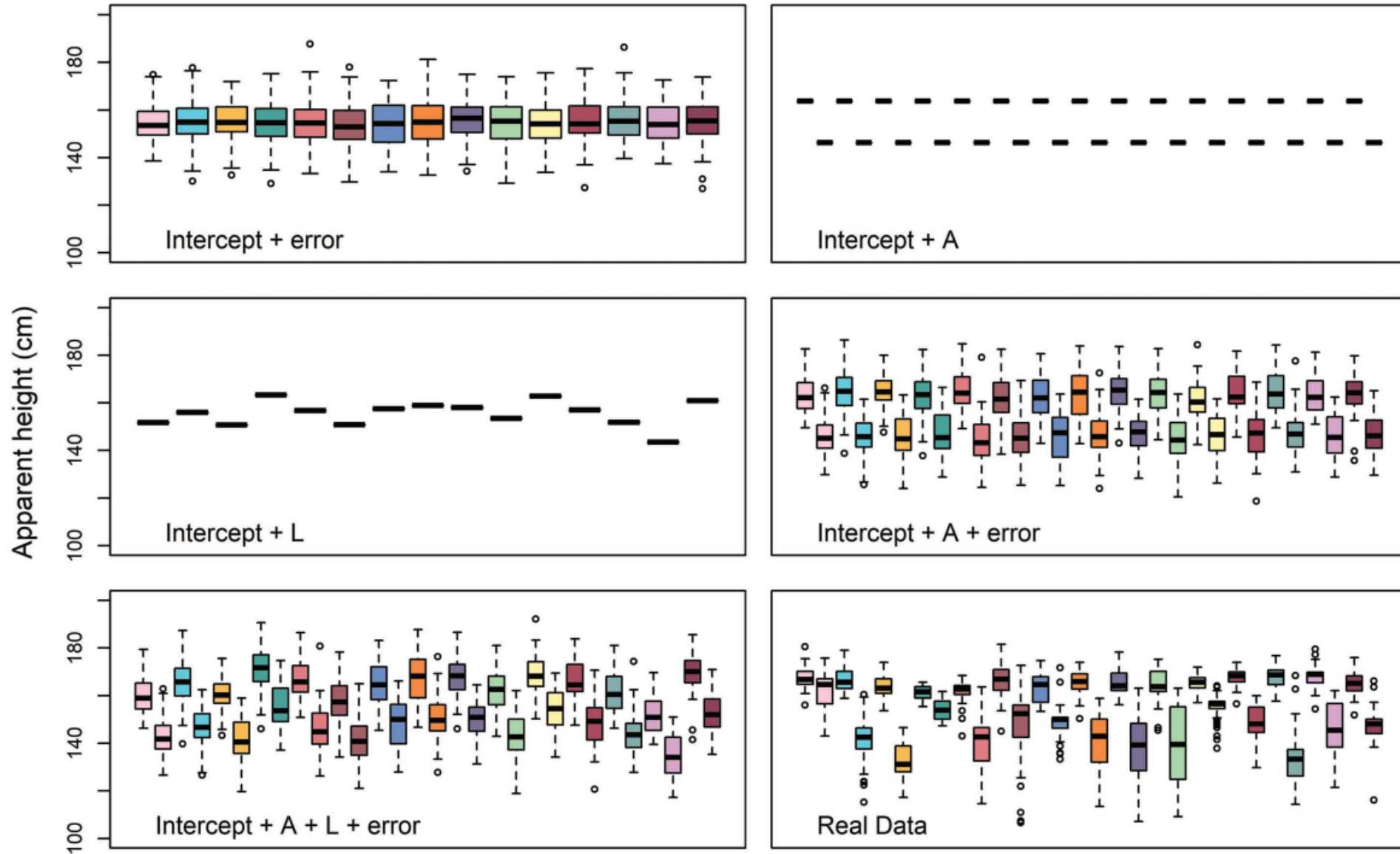
```
# this is a vector of 94 speaker effects
S_ = rnorm(n_speakers, 0, 3.6)
```

```
# this is a vector indicating the speaker
S = rep(1:n_speakers, each = n_listeners)
```

```
# this vector contains the error
epsilon = rnorm(n_speakers*n_listeners, 0, 8.6)
```

```
# the sum of the above components equals our observations
height_rep = Intercept + A_[A] + L_[L] + S_[S] + epsilon
```

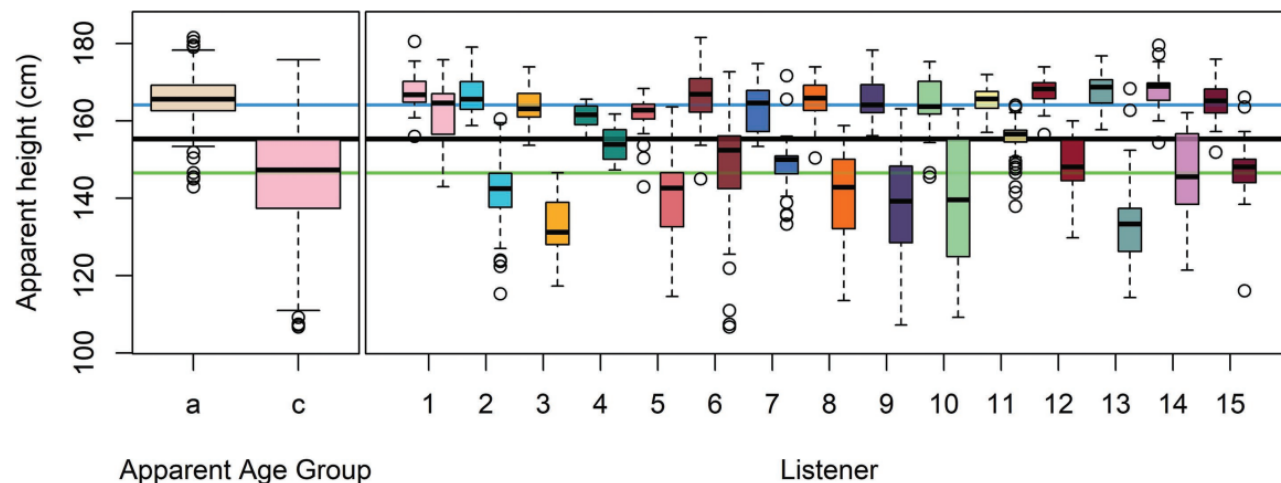
Simulating Two-Group Data



Answering Our Research Questions

- (1) ¿Qué tan altos suenan los hablantes percibidos como mujeres adultas?
- (2) ¿Qué tan altos suenan los hablantes percibidos como niños?
- (3) ¿Cuál es la diferencia de altura aparente asociada a la percepción de la adultez?

The overall mean apparent height across all speakers was 156 cm (s.d. = 1.38, 95% C.I. = [152.85, 158.31]). We found a difference of 17 cm (s.d. = 0.68, 95% C.I. = [15.56, 18.24]) in apparent height associated with the perception of adulthood in speakers. The standard deviation of the listener and speaker effects were 5.1 cm (s.d = 1.1, 95% CI = [3.5, 7.8]) and 3.4 cm (s.d = 0.4, 95% CI = [2.6, 4.2]) respectively. Overall, results indicate a reliable difference in apparent speaker height due to apparent age, which is larger than the expected random variation in apparent height judgments due to variation between speakers and listeners.



Traditionalist's Corner: lmer

```
library (lme4)
lmer_model = lmer (height ~ A + (1|L) + (1|S), data = notmen)
```

```
summary (lmer_model)
```

```
## Linear mixed model fit by REML ['lmerMod']
```

```
## Formula: height ~ A + (1 | L) + (1 | S)
```

```
## Data: notmen
```

```
##
```

```
## REML criterion at convergence: 10161
```

```
##
```

```
## Random effects:
```

```
## Groups Name Variance Std.Dev.
```

```
## S (Intercept) 12.4 3.52
```

```
## L (Intercept) 22.9 4.79
```

```
## Residual 73.5 8.57
```

```
## Number of obs: 1401, groups: S, 94; L, 15
```

```
##
```

```
## Fixed effects:
```

```
## Estimate Std. Error t value
```

```
## (Intercept) 155.193 1.310 118.5
```

```
## A1 8.729 0.312 27.9
```

```
##
```

```
## Correlation of Fixed Effects:
```

```
## (Intr)
```

```
## A1 0.046
```

```
fixef (model_sum_coding)
```

```
## Estimate Est.Error Q2.5 Q97.5
```

```
## Intercept 155.218 1.4050 152.434 157.965
```

```
## A1 8.711 0.3602 8.003 9.416
```

Exercises

Use the data in 'exp_ex' to do one of the following. You may also use your own data to answer a related question. In any case, describe the model, present and explain the results, and include at least two figures.

1. **Easy:** Analyze the (pre-ft) model that's exactly like `model_sum_coding_t`, except using the data in `exp_ex` (`bmmmb::get_model("5_model_sum_coding_t_ex.RDS")`).
2. **Medium:** Fit a model like `model_sum_coding_t`, but comparing any two groups across resonance levels.
3. **Hard:** Fit two models like `model_sum_coding_t`, but compare any two groups across resonance levels. Compare results across models to think about group differences.